



AI-Driven Multi-Modal Monitoring System for Crop Health, Soil Analysis, and Pest Risk Detection

TEAM PRITHVI





PROBLEM STATEMENT

Crop stress, soil erosion, and pests infestation are major factors that affect agricultural productivity leading to huge economic and food security problems in the world. The Food and Agriculture Organization (FAO) estimated that annually 20–40% of the global crop production is lost to pests and diseases, which leads to about 220 billion dollars of economic losses annually, and nearly 33% of soils on the planet are in moderate or severe conditions of degradation which lowers yield potential significantly. The climate change also enhances the issue, with the higher temperatures and humidity promoting the quick reproduction of pests and their increase in the frequency of outbreaks. Current monitoring systems are either isolated- satellite platforms can only observe vegetation indices, the nutrients in the soil are monitored by the soil sensor without the context of the crop and image-based disease detection can only notice the problem after symptoms have been observed. Such disjointed and reactive strategies cannot deliver the early, predictive and holistic agricultural intelligence, which demonstrates why there is an urgent need for an AI-based system that integrates crop imagery, soil data, and real-time weather intelligence to enable predictive and proactive agricultural monitoring.





EXISTING SYSTEMS

IOT-BASED SOIL MONITORING SYSTEMS

Sensor-based systems measure soil moisture, pH, and nutrient levels to assist irrigation and fertilizer management

WEATHER-BASED ADVISORY SYSTEMS

Weather-based agricultural advisory systems use meteorological data such as temperature, humidity, rainfall, and wind patterns to provide irrigation planning, sowing recommendations, and pest outbreak warnings.

IMAGE-BASED DISEASE DETECTION SYSTEMS

Deep learning models analyze RGB leaf images to detect visible crop diseases.



LIMITATIONS

- Lack of multi-modal data integration (image, soil, and weather processed separately)
- Limited real-time data synchronization and streaming capability.
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- Mostly reactive analytics without predictive modeling.
- Limited interoperability between different agricultural data platforms.





PROPOSED SOLUTION – CORE IDEA



RGB CROP IMAGES (USER INPUT)



Soil parameter analysis (pH, NPK, moisture)

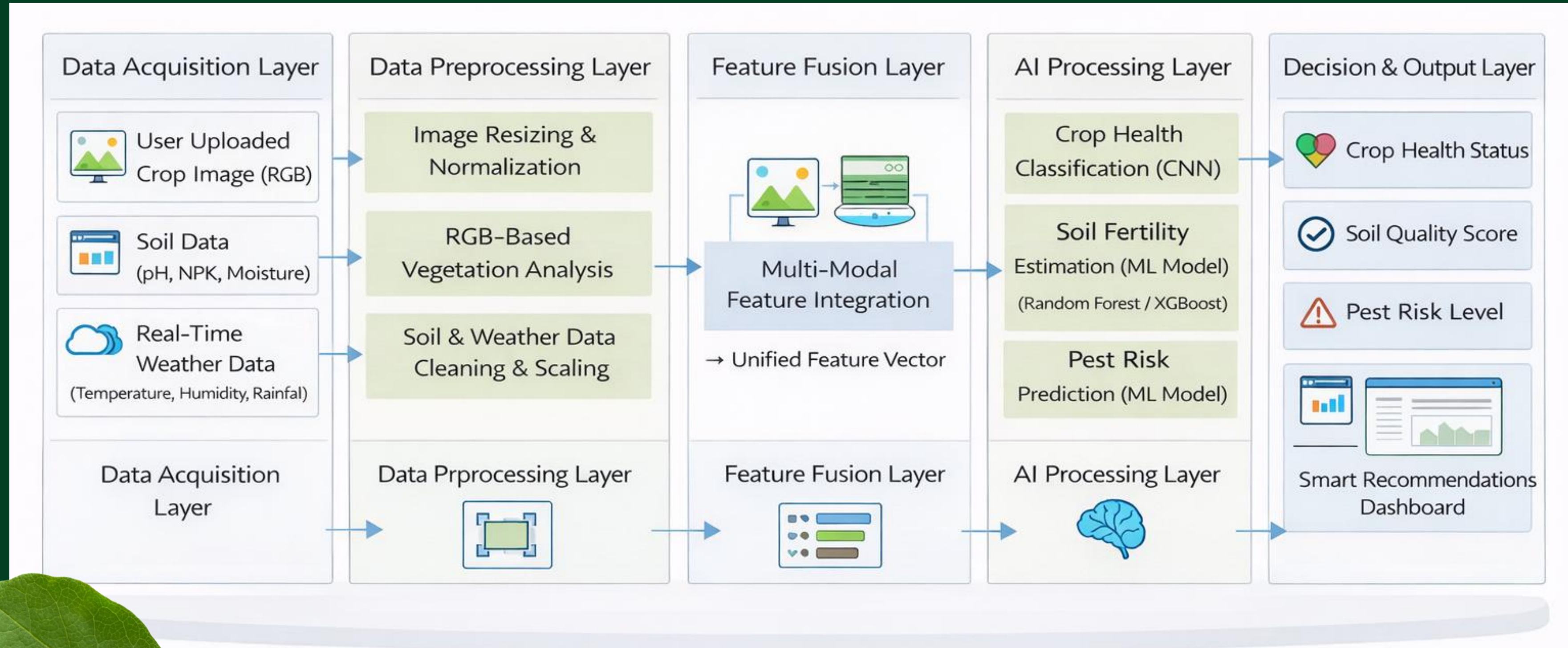


Weather-based pest risk prediction

In our proposed system, RGB crop images uploaded by the user are combined with soil parameters (pH, NPK, moisture) and real-time weather data. These inputs are preprocessed and fused into a unified feature vector. Machine learning and deep learning models then classify crop health, estimate soil fertility, and predict pest risk levels. By integrating crop imagery with environmental intelligence, the system enables predictive and proactive agricultural monitoring. The resultant fused dataset is subsequently analyzed by machine learning, deep learning models to classify crop health, estimate soil fertility and already predict pest risk levels. With integration of weather parameters and crop and soil intelligence, the system allows early detection and proactive action instead of identifying problems when they arise.



SYSTEM ARCHITECTURE





Thank You!

Agriculture continues to evolve, feeding the world while adapting to new challenges. Thank you for your time.

